

Code-Layout, Readability and Reusability

Date: 05 October 2026

Team ID: SWTID-2026-8809

Project Name: EduGenie: Google Gemini Powered Learning Assistant

Maximum Marks: 5 Marks

Code-Layout, Readability and Reusability

This section evaluates the EduGenie codebase in terms of structure, readability and reusability. The project uses separate modules for the FastAPI application, explanation, Q&A, quiz, summary and learning-path functionality, with HTML/CSS frontend files.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Maintain consistent formatting.
2	Proper File Structure	Files and folders are logically organized.	Yes	Separate backend modules and frontend folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly.	Yes	Use descriptive names.
4	Function / Method Names	Functions are descriptively named.	Yes	Feature-specific function names are used.
5	Code Comments	Inline and block comments explain logic.	Partial	Add comments where implementation needs clarification.
6	Modular Design	Code is split into reusable functions/modules.	Yes	Learning features are separated into modules.
7	No Redundant Code	Duplicate or unused code is removed.	Yes	Review during final integration.
8	Error Handling	Exceptions and errors are handled gracefully.	Yes	Response and parsing errors are handled.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Level
1	FastAPI application	Python / FastAPI	Backend API endpoints	High
2	Explanation Module	Python / LaMini-Flan-T5	Concept explanation	High
3	Q&A Module	Python / Gemini	Question answering	High
4	Quiz Module	Python / Gemini	Quiz generation	High
5	Summary Module	Python / Gemini	Text summarization	High
6	Learning Path Module	Python / Gemini	Learning recommendations	High
7	HTML/CSS Frontend	HTML / CSS	User interface	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Modular backend and frontend structure.
Readability	4	Clear module separation and descriptive feature names.
Reusability	5	Core learning functions are separated into reusable modules.
Documentation / Comments	4	Documentation is available; comments can be expanded.
Overall Score	4.5 / 5	Good structure, readability and modularity.

Team Work Allocation

S.No	Team Member	Assigned Work
1	Divya K	Overall code structure and architecture; file organization, integration and final code-quality review.
2	Veronica	Q&A and Explanation modules; readable functions, naming, comments and error handling.
3	Sangeetha T	Quiz and Summary modules; response handling, JSON parsing and reusable functions.
4	Pooja N	Learning Path module and FastAPI endpoints; API structure, error handling and backend reusability.
5	Monika N	HTML/CSS frontend; readable layout, reusable UI structure, styling and integration.

Team Members: Divya K | Veronica | Sangeetha T | Pooja N | Monika N